Assessment of Low-Carbon Technologies in the Iron and Steel Industry

Tsinghua University & Saudi Aramco

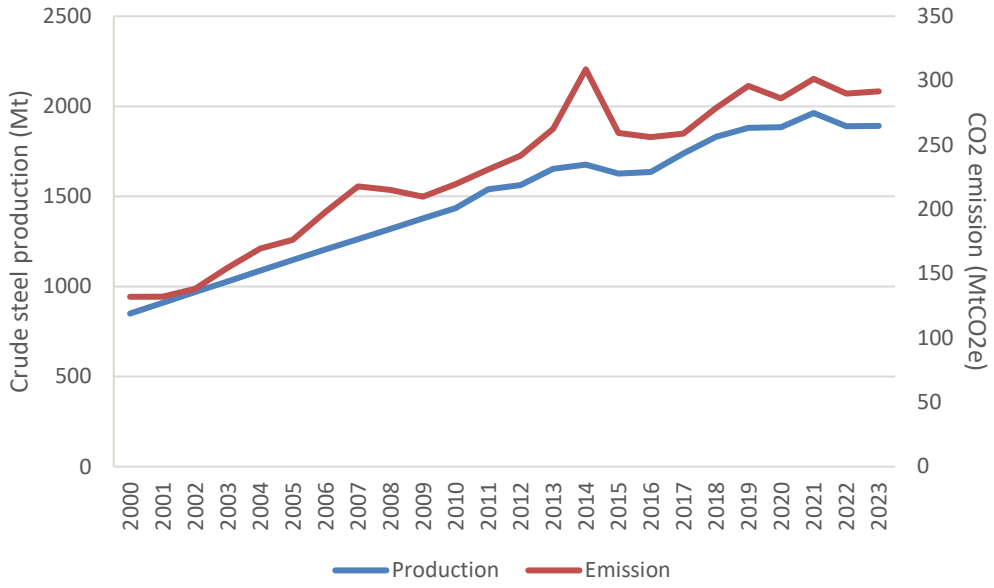
aramco

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

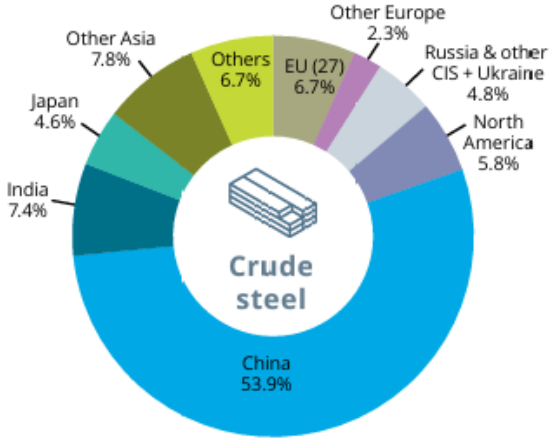
Presenter: Yunchuan Liu, PhD student

August 11, 2025

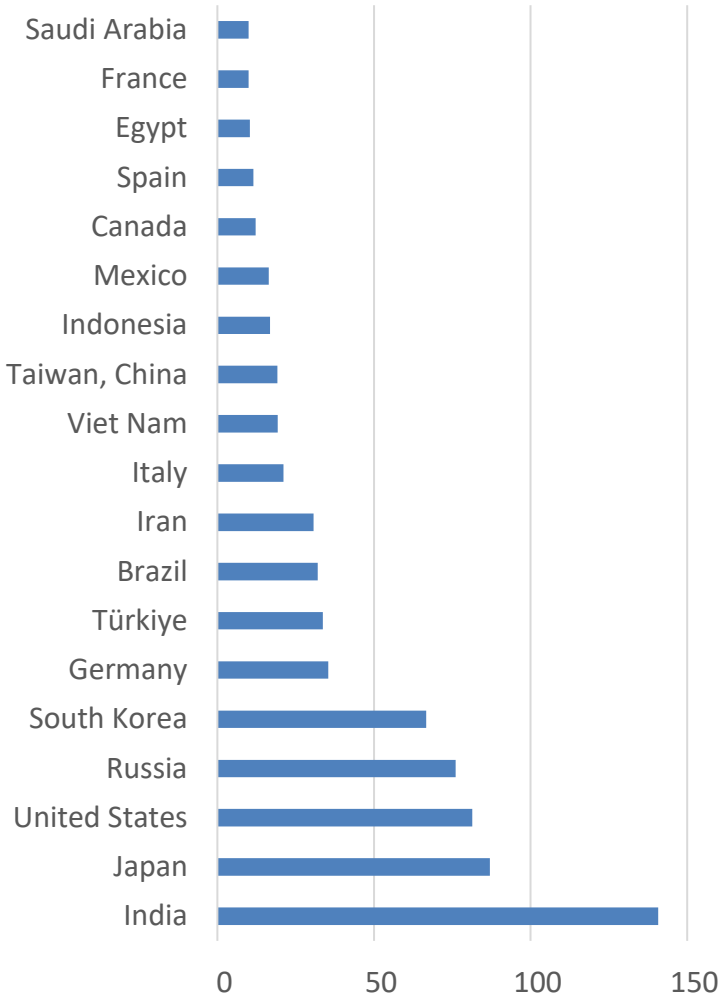
The iron & steel industry is a key area of global emission reduction



World crude steel production & emission



Geographical distribution of steel production



Top 20 regions' steel production in 2023 (Mt, except China Mainland)

- The scale and carbon emissions of the steel industry have increased rapidly in recent years, the iron & steel industry accounts for approximately 7-9% of global carbon dioxide emissions.
- Nearly 20 countries outside China have steel production capacity of 10 million tons or close to it. Their steel production emissions reduction also requires attention.

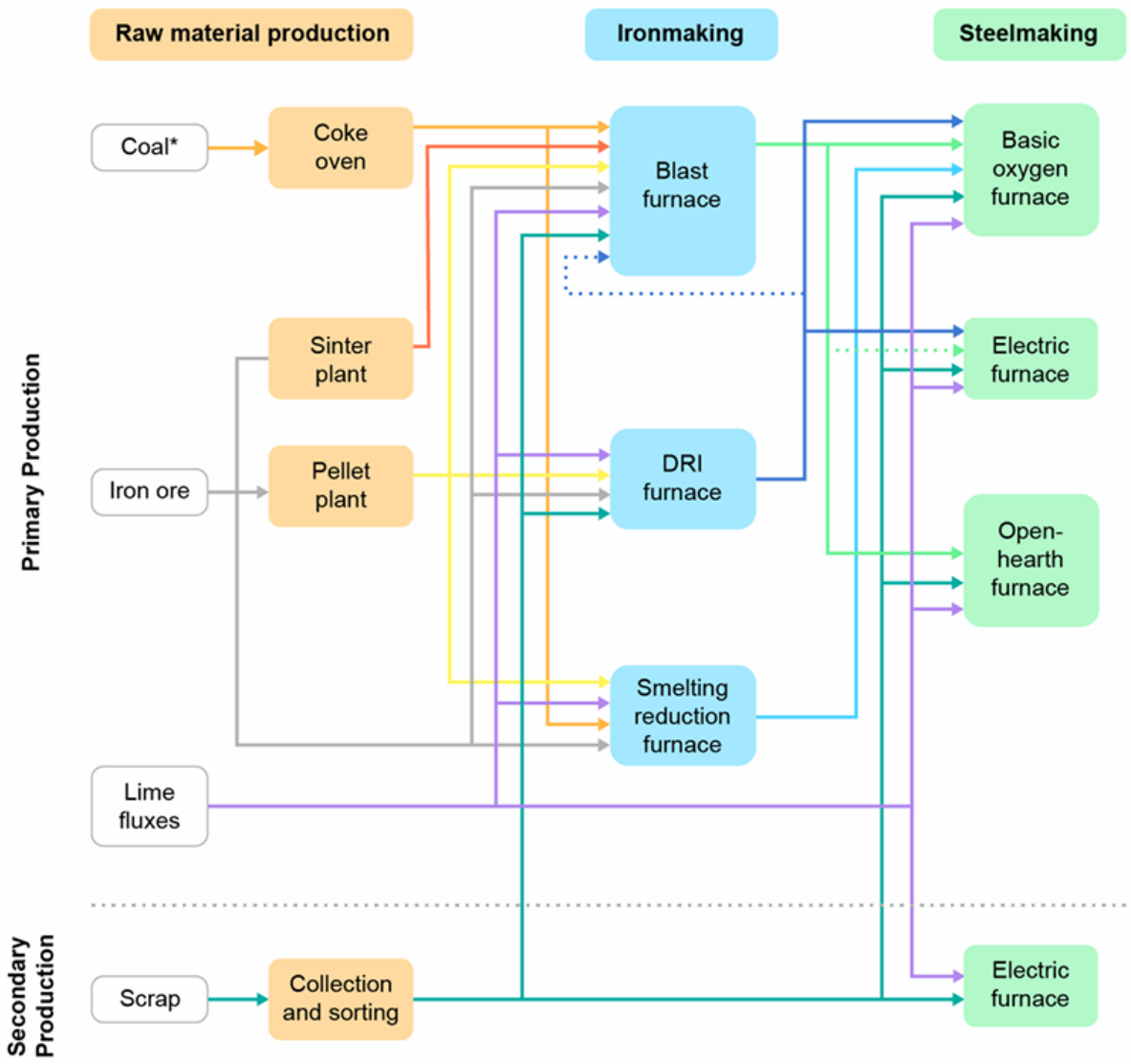
Source: IEA, EDGAR, Worldsteel

Main technical routes of the worldwide steel industry

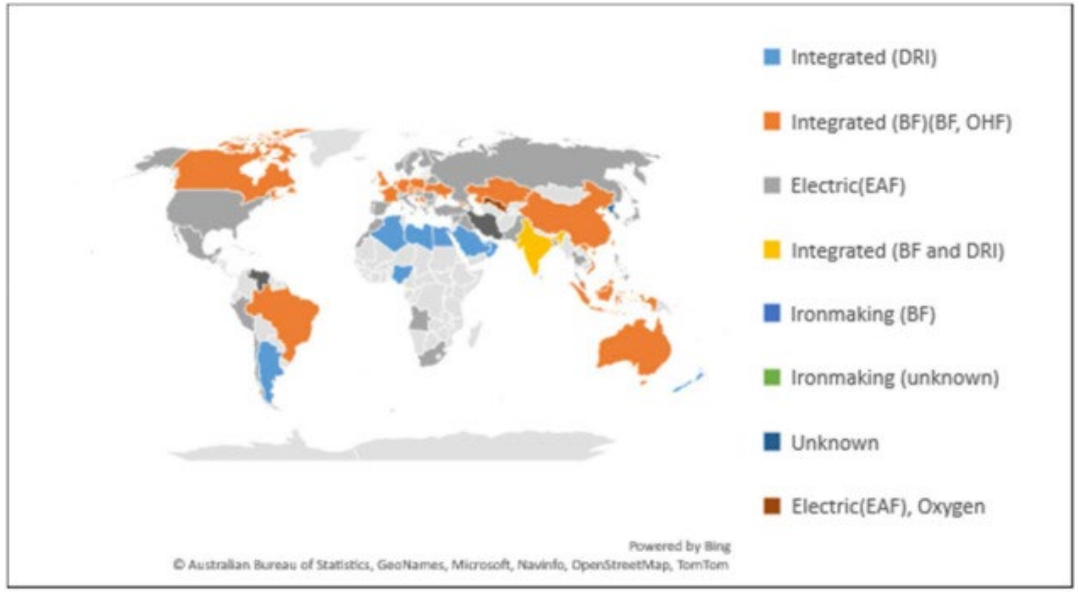
Technology process	reductant	CO2 emission per crude steel (tCO_2/tcs)	Share of global steel production (%)
BF-BOF	Coke	1.7-2.5	73.2
DRI-EAF	NG	1.5-1.7	3.8
	Coal	2.5-3.2	1.0
	Hydrogen	/	/
Scrap-EAF	/	0.35-0.66	21.5
Others (BF-OHF, etc.)			

- BF-BOF, Scrap-EAF and DRI-EAF are currently the dominant steel production technology routes.
- Coal-based DRI-EAF routes and traditional BF-BOF routes have the highest carbon emissions, while gas-DRI-EAF has the lowest carbon emission in long process steelmaking routes.
- **H2-DRI-EAF (green hydrogen required)** is often considered as a potential technical route for deep decarbonization.

Source: IEEFA, https://ieefa.org/sites/default/files/2022-06/steel-fact-sheet.pdf



Worldwide steel production technology structure

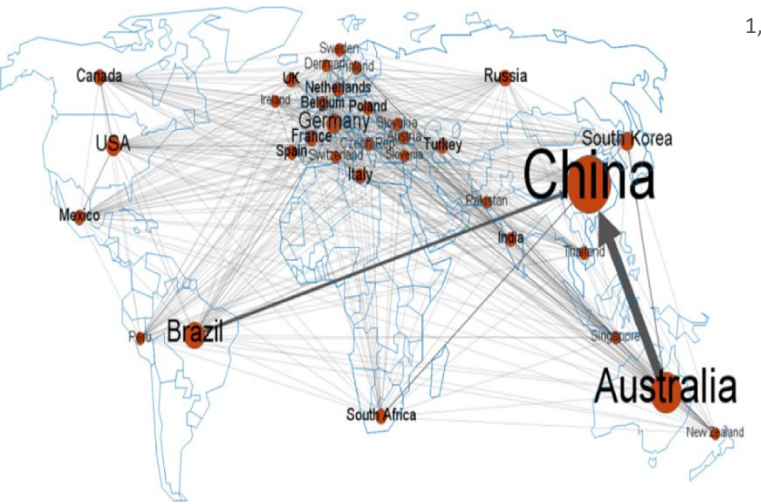


Color map of steel production technology structure

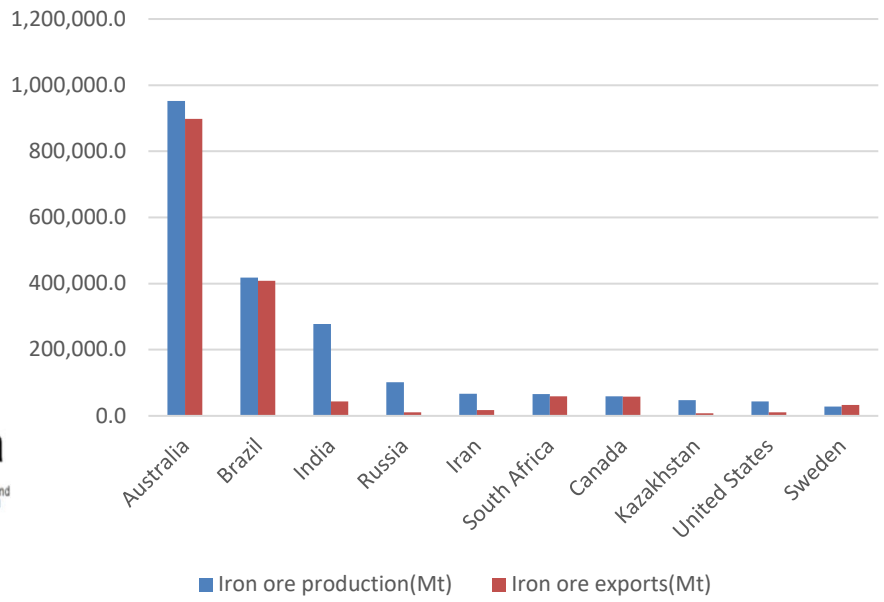
Example country	Technology mix		DRI Production (Mt)
	BOF rate(%)	EAF rate(%)	
Saudi Arabia	0	100	6.8
Austrailia	73.2	26.8	/
Japan	73.8	26.2	/
USA	31.7	68.3	5.2
India	43.6	56.4	49.6
Germany	72.3	27.7	0.2
Russia	65.1	32.0	7.8

- There are significant differences in the steel production technology structure of different countries, which is caused by a combination of factors such as energy prices, resource endowments, and industrial policies.
- Most countries mainly use the BF-BOF technology route, while a few countries mainly use the Scrap-EAF or DRI-EAF technology route.

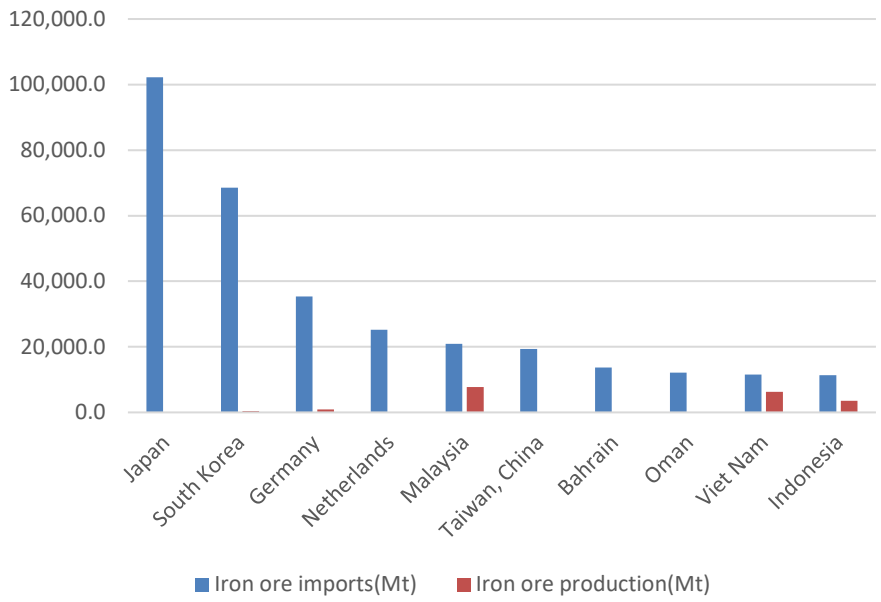
International division of labor in steel industry



International iron ore trade flow



Iron ore production & exports in selected countries (Mt, 2023, Top10 iron ore producing countries)



Iron ore import & production in selected countries (Mt, 2023, Top10 iron ore import countries)

- The steel industry is highly internationalized, with iron ore trade being a key component.
- There are countries where the mining industry depends on iron ore exports, or the steel industry depends on iron ore imports.

Zhong, Weiqiong, et al. "Structure of international iron flow: based on substance flow analysis and complex network." Resources, Conservation and Recycling 136 (2018): 345-354.

A brief introduction to different types of DRI products

DRI Product	Manufacture	Transport	Storage
HDRI	The **direct product** formed after iron ore is reduced	● Localized transport only	● Not applicable
CDRI	The product formed by the **direct cooling** of HDRI	● Water (Inland): Must be protected from rain & moisture ● Water (Ocean): Holds must be kept inert per IMO guidelines ● Rail & Road: Must be protected from rain & moisture	● Silos & Bins: Must be kept cool & dry ● Piles: Must be covered with proper drainage
HBI	The product formed by **briquetting HDRI** (under pressure) at a high temperature of 650° C	● Water (Inland): No special precautions ● Water (Ocean): No special precautions ● Rail & Road: No special precautions	● Silos & Bins: No special precautions ● Piles: Uncovered with good drainage

- Once DRI products are processed into CDRI or HBI, they can be transported and traded.
- DRI products contain more Fe than iron ore, and trading DRI instead of iron ore may reduce transportation costs and carbon emissions.

Source: https://www.midrex.com/wp-content/uploads/MIdrexDRI_ProductsBrochure_4-12-18.pdf

Research questions

Q1： What are the economic and emission reduction benefits of H2-DRI-EAF in the world's major steel-producing countries?

- **Different case countries**
  - iron ore resource endowments
  - energy costs, labor costs, hydrogen production costs
  - market environments, steel demand, ...
- **A study about technology suitability which need more case studies is required**

Q2： If DRI trade is used to replace iron ore trade, can it effectively reduce the costs and emissions of the steel production chain?

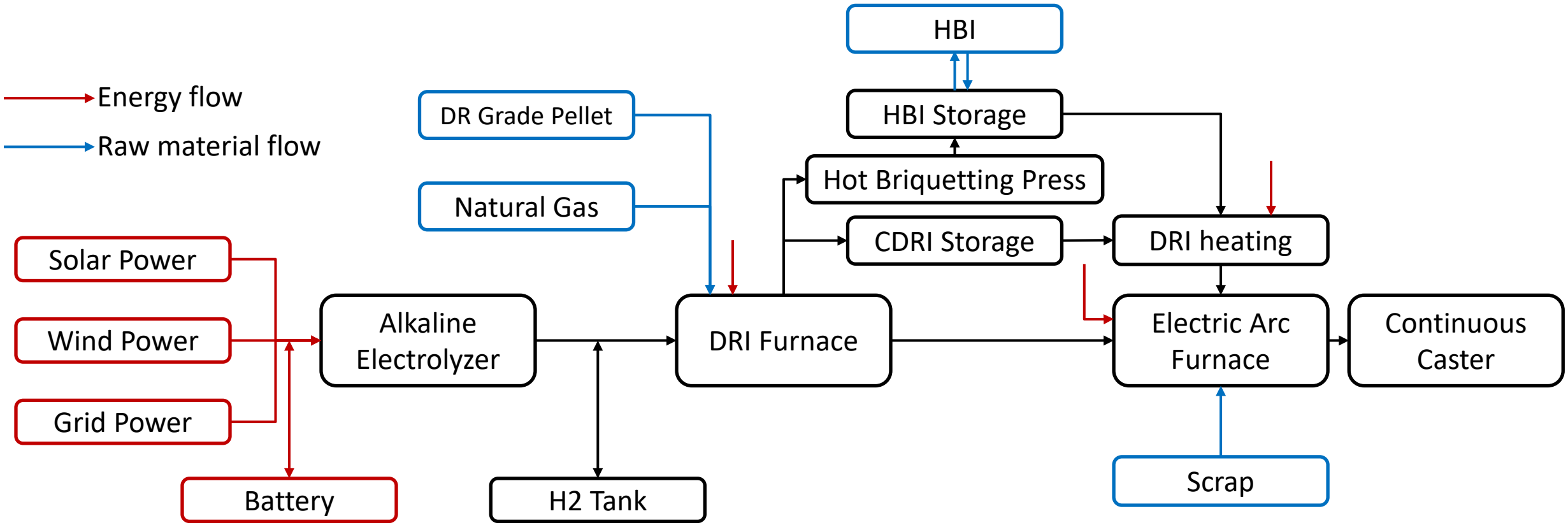
- **Identify typical iron ore trade routes**
  - Situation in exporting and importing countries
  - Transport distance, ...
- **A study about Economic benefits of using HBI trade instead of iron ore trade is required**

Methodology

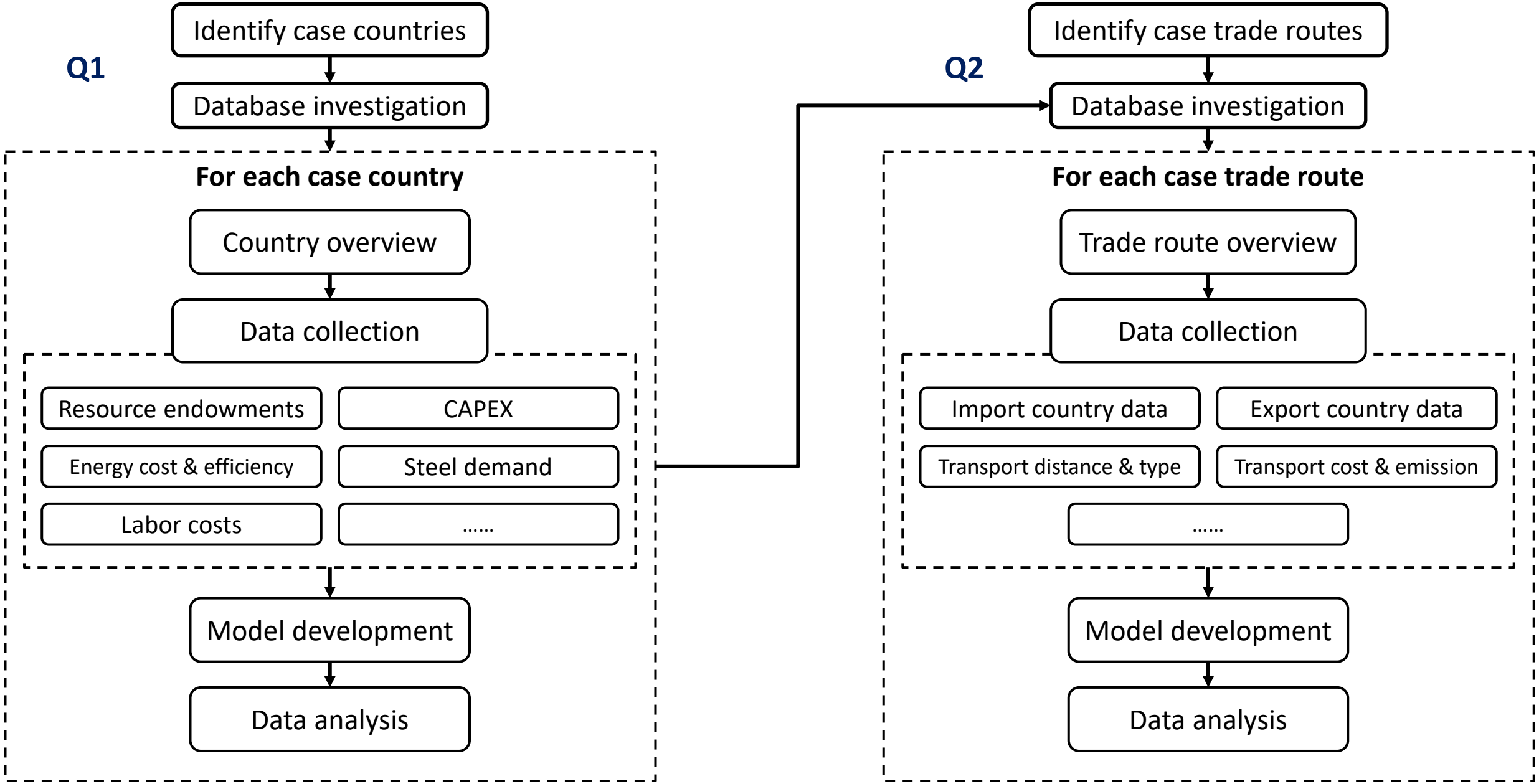
H2-DRI-EAF production scheduling model with flexibility

Hourly resolution system optimization model

Consider the **production, storage and consumption** of **electricity, hydrogen, iron and steel**



Research roadmap & Data requirements



Case selection

Country	Region	Crude steel production (Mt)	Iron ore production (Mt)	BOF rate(%)	EAF/IF rate(%)
Saudi Arabia	Middle East	9.9(20th)	0.0	0	**100**
Austrailia	Oceania	5.5(28th)	944.1(1st)	**73.2**	26.8
Japan	Asia	87.0(3rd)	0.0	**73.8**	26.2
USA	North America	81.4(4th)	46.0	31.7	**68.3**
India	Asia	140.8(2nd)	247.8(4th)	**43.6**	**56.4**
Germany	Europe	35.4(7th)	0.8	**72.3**	27.7
Russia	Europe	76.0(5th)	90.9	**65.1****	32.0**

- Seven countries with relatively high steel production are selected
- The selection also ensures diversity in terms of geography, technological structure and international division of labor.

Year: **2023**; Data sources: **Worldsteel, IRENA, World Bank, Rosstat**
* Worldsteel provided data for Russia & other CIS + Ukraine: ore production volume 174.8 Mt, exports 44.1 Mt, imports 0; Rosstat provided Russia's ore production volume as 90.9 Mt; figures were derived by proportional scaling.
** Russia still has open-hearth furnace steelmaking, which accounts for the remaining 2.9% of its technological structure.
*** Wind power installation cost data is provided based on the country's region.

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Database: DRI location

- steelonthenet.com (SOTN): Proprietary
  - Plant-level capacity data, technology & location specific, build-up and retirement year (for each unit)
  - Scope: Crude steel , upstream (coke, sinter, pig iron/DRI, crude steel), location

18 JANUARY 2024
Table
IRON AND STEEL CAPACITY BY PLANT AND BY COMPANY - ALL PRODUCTS, PLANTS AND COUNTRIES
(000 tonnes per year capacity at 31 December 2023, ownership at date of update)

Owner	Owner Name	Owner Country	Site	Plant Region	Plant Country	Plant Name	Plant Location		coke	sinter	pig iron			DRI	crude steel			
											BF iron	other pig iron	total pig iron		EAF steel	BOF steel	OHF steel	total crude
			3889	Asia - Middle East	Oman	Mill	Sohar	..					0.0	169.0	742.9			742.9
			15629	Asia - Middle East	Oman	Sun Metals	Sur	..					0.0		2562.5			2562.5
			3890	Asia - Middle East	Qatar	Qasco - Qatar Steel	Umm Said	..					0.0	2635.0	3758.8			3758.8
			16026	Asia - Middle East	Saudi Arabia	Al-Atoun Steel Industries	Yanbu	..					0.0		932.8			932.8
			286	Asia - Middle East	Saudi Arabia	Al-Azizia Steel	Hadda	..					0.0		522.4			522.4
			15547	Asia - Middle East	Saudi Arabia	Al-Qaryan Steel	Dammam	..					0.0		309.0			309.0
			14142	Asia - Middle East	Saudi Arabia	Al-Rahji Steel	Jeddah, Alkhumra	..					0.0		976.9			976.9
			3892	Asia - Middle East	Saudi Arabia	Al-Tuwairqi Steel	Dammam	..					0.0	**1706.9**	1134.5			1134.5
			13267	Asia - Middle East	Saudi Arabia	ArcelorMittal Tubular Products	Jubail	..					0.0		649.6			649.6
			3895	Asia - Middle East	Saudi Arabia	Hadeed - Saudi Iron & Steel	Jubail	..					0.0	**6854.3**	8986.2			8986.2
			14891	Asia - Middle East	Saudi Arabia	Hill Metals	Jeddah	..					0.0		132.5			132.5
			13942	Asia - Middle East	Saudi Arabia	South Steel - Solb	Jizan Economic City	..					0.0		1082.7			1082.7
			14892	Asia - Middle East	Saudi Arabia	Watania Steel	Riyadh	..					0.0		318.4			318.4
			3913	Asia - Middle East	Syria	Gecosteel	Hama	..					0.0		245.2			245.2
			15952	Asia - Middle East	Syria	Hmisho Steel	Latakia	..					0.0		563.0			563.0
			3900	Asia - Middle East	United Arab Emirates	Ahli Steel	Dubai	D					0.0		522.4			522.4
			15746	Asia - Middle East	United Arab Emirates	Arabian Gulf Steel Industries	not known	..					0.0		424.5			424.5
			3914	Asia - Middle East	United Arab Emirates	ESI - Emirates Steel Industries	Mussafah Industrial City A	Al					0.0		342.4			342.4
			3901	Asia - Middle East	United Arab Emirates	ESI - Emirates Steel Industries	Mussafah Industrial City B	Al					0.0	2122.7	3608.6			3608.6
			14872	Asia - Middle East	United Arab Emirates	Euro Gulf Steel Industries	Dubai	D					0.0		401.6			401.6
			15748	Asia - Middle East	United Arab Emirates	Taybah Steel	Fujairah	Fu					0.0		424.5			424.5
			15406	Asia - Middle East	Yemen	Mukalla Iron and Steel	not known	..					0.0		165.7			165.7

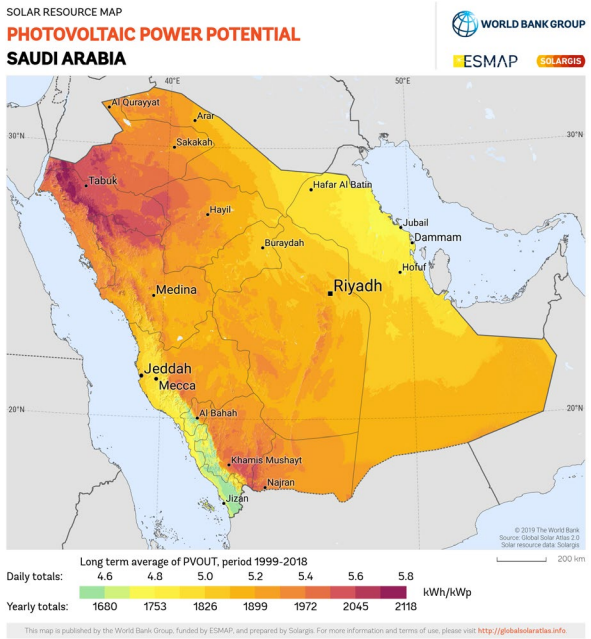
Database: Maximum Renewable Output

- Renewables.ninja : Open Source
  - Maximum Solar PV Output, calculated from MERRA-2 weather dataset (2019)
  - Maximum Wind power Output, calculated from MERRA-2 weather dataset (2019)
  - Scope: Worldwide, hourly data in 2019

# Renewables.nir	49.653 - Version: 1.1 (using	electricity ir
# Units: time in U	local_time in Asi	
# {"units": {"time"	"local_time": "As	"electricity
time	local_time	electricity
2019/1/1 0:00	2019/1/1 3:00	0
2019/1/1 1:00	2019/1/1 4:00	0
2019/1/1 2:00	2019/1/1 5:00	0
2019/1/1 3:00	2019/1/1 6:00	0.006
2019/1/1 4:00	2019/1/1 7:00	0.196
2019/1/1 5:00	2019/1/1 8:00	0.419
2019/1/1 6:00	2019/1/1 9:00	0.579
2019/1/1 7:00	2019/1/1 10:00	0.673
2019/1/1 8:00	2019/1/1 11:00	0.706
2019/1/1 9:00	2019/1/1 12:00	0.67
2019/1/1 10:00	2019/1/1 13:00	0.551
2019/1/1 11:00	2019/1/1 14:00	0.402
2019/1/1 12:00	2019/1/1 15:00	0.136
2019/1/1 13:00	2019/1/1 16:00	0.022
2019/1/1 14:00	2019/1/1 17:00	0
2019/1/1 15:00	2019/1/1 18:00	0
2019/1/1 16:00	2019/1/1 19:00	0
2019/1/1 17:00	2019/1/1 20:00	0
2019/1/1 18:00	2019/1/1 21:00	0
2019/1/1 19:00	2019/1/1 22:00	0
2019/1/1 20:00	2019/1/1 23:00	0
2019/1/1 21:00	2019/1/2 0:00	0
2019/1/1 22:00	2019/1/2 1:00	0
2019/1/1 23:00	2019/1/2 2:00	0
2019/1/2 0:00	2019/1/2 3:00	0
2019/1/2 1:00	2019/1/2 4:00	0
2019/1/2 2:00	2019/1/2 5:00	0

# Renewables.ninj	49.653 - Version: 1.1 - Licen	
# Units: time in U	local_time in Asia	electricity i
# {"units": {"time":	"local_time": "Asia	"electricity
time	local_time	electricity
2019/1/1 0:00	2019/1/1 3:00	0.449
2019/1/1 1:00	2019/1/1 4:00	0.394
2019/1/1 2:00	2019/1/1 5:00	0.345
2019/1/1 3:00	2019/1/1 6:00	0.319
2019/1/1 4:00	2019/1/1 7:00	0.261
2019/1/1 5:00	2019/1/1 8:00	0.174
2019/1/1 6:00	2019/1/1 9:00	0.108
2019/1/1 7:00	2019/1/1 10:00	0.07
2019/1/1 8:00	2019/1/1 11:00	0.03
2019/1/1 9:00	2019/1/1 12:00	0.016
2019/1/1 10:00	2019/1/1 13:00	0.029
2019/1/1 11:00	2019/1/1 14:00	0.068
2019/1/1 12:00	2019/1/1 15:00	0.091
2019/1/1 13:00	2019/1/1 16:00	0.134
2019/1/1 14:00	2019/1/1 17:00	0.165
2019/1/1 15:00	2019/1/1 18:00	0.13
2019/1/1 16:00	2019/1/1 19:00	0.068
2019/1/1 17:00	2019/1/1 20:00	0.021
2019/1/1 18:00	2019/1/1 21:00	0.006
2019/1/1 19:00	2019/1/1 22:00	0.007
2019/1/1 20:00	2019/1/1 23:00	0.022
2019/1/1 21:00	2019/1/2 0:00	0.044
2019/1/1 22:00	2019/1/2 1:00	0.077
2019/1/1 23:00	2019/1/2 2:00	0.112
2019/1/2 0:00	2019/1/2 3:00	0.179
2019/1/2 1:00	2019/1/2 4:00	0.223
2019/1/2 2:00	2019/1/2 5:00	0.278
2019/1/2 3:00	2019/1/2 6:00	0.279
2019/1/2 4:00	2019/1/2 7:00	0.212

Hourly PVOut & Wind capacity factor data - Jubail & Dammam, Saudi-Arabia



PVOut - Saudi Arabia(kWh/kWp)

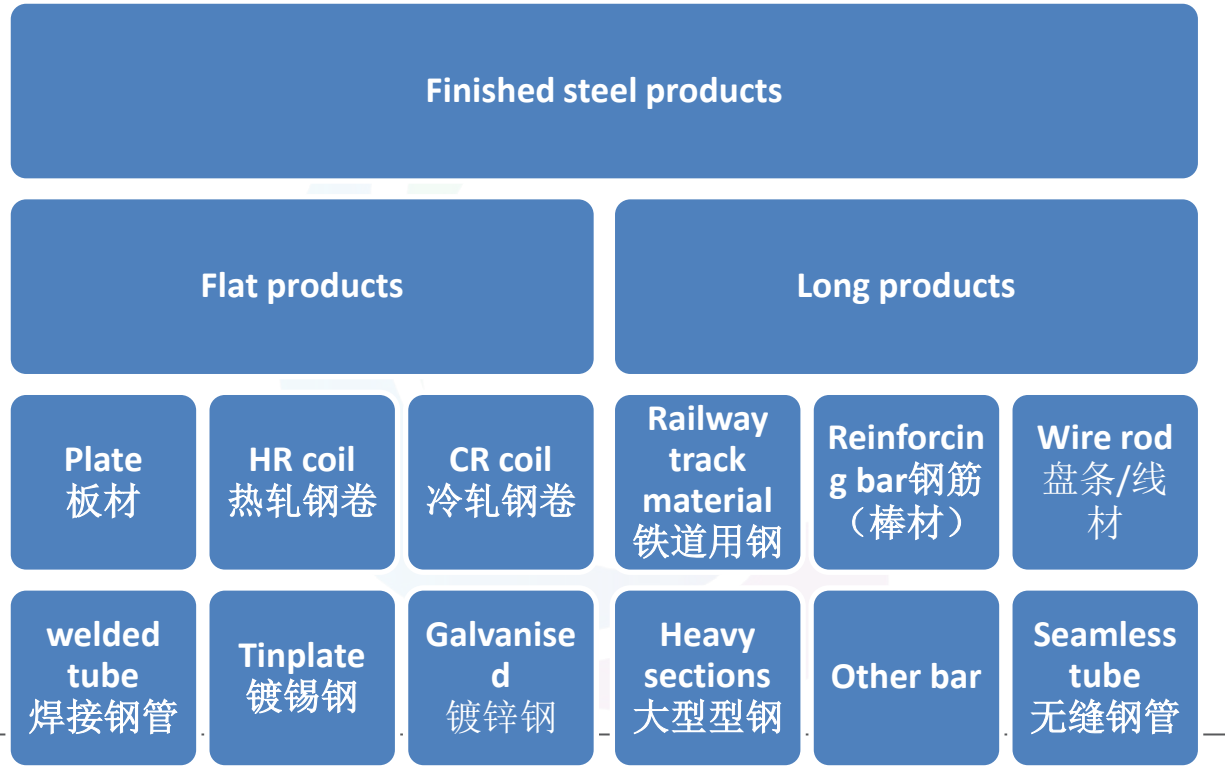
Capacity factor - Saudi Arabia(kWh/kWp)
from: Global wind atlas

Database: Steel demand

- steelonthenet.com (SOTN): Proprietary
  - Consumption data from 1970~2050, including predictions, national-level
  - Import & export data from 1970~2050

FINISHED STEEL CONSUMPTION - TOTAL
(000 tonnes)

Region	Country	Product	Item	16	2017	2018	2019	2020	2021	2022	2023
North America	Canada	Finished stee	Consumption	.7	12725.4	15377.9	15951.1	13166.0	13604.2	14888.5	14092.4
North America	United States	Finished stee	Consumption	.5	97380.2	99399.1	97284.2	80156.0	99275.5	87410.5	91841.9
Latin America	Argentina	Finished stee	Consumption	.0	5002.6	4639.0	4144.2	3528.6	5019.4	3868.5	2875.7
Latin America	Bolivia	Finished stee	Consumption	.2	735.9	721.0	736.0	509.9	804.0	570.6	559.2
Latin America	Brazil	Finished stee	Consumption	.7	21970.3	24554.6	23081.2	22672.6	28839.1	25464.0	25765.4
Latin America	Chile	Finished stee	Consumption	.9	2668.6	3076.4	2935.8	2455.9	3306.7	1981.2	1961.9
Latin America	Colombia	Finished stee	Consumption	.0	4121.8	4172.6	4131.5	3053.5	4124.6	3979.7	3171.9
Latin America	Costa Rica	Finished stee	Consumption	.3	736.3	788.5	697.2	561.3	743.2	665.5	700.3
Latin America	Cuba	Finished stee	Consumption	.0	299.4	269.0	268.8	150.5	167.9	194.5	216.0
Latin America	Dominican Re	Finished stee	Consumption	.2	775.3	853.1	959.0	751.2	983.0	602.0	678.3
Latin America	Ecuador	Finished stee	Consumption	.1	1650.1	1657.5	1665.5	1264.2	1721.1	1606.6	1667.2
Latin America	El Salvador	Finished stee	Consumption	.6	405.0	420.1	417.8	370.4	549.4	338.5	378.6
Latin America	Guatemala	Finished stee	Consumption	.7	844.9	831.9	932.6	851.7	1177.4	880.7	961.5
Latin America	Honduras	Finished stee	Consumption	.9	433.4	473.9	436.4	373.3	544.0	178.6	212.0
Latin America	Jamaica	Finished stee	Consumption	.5	176.2	151.4	198.5	174.2	237.4	85.3	83.2
Latin America	Mexico	Finished stee	Consumption	.4	25796.2	26275.5	24221.4	21799.0	26085.0	23650.8	23957.2
Latin America	Nicaragua	Finished stee	Consumption	.7	280.7	212.6	249.3	280.2	364.1	183.2	201.1
Latin America	Panama	Finished stee	Consumption	.9	514.4	460.5	378.7	169.7	338.7	342.7	338.8
Latin America	Paraguay	Finished stee	Consumption	.5	358.3	446.2	348.0	302.7	388.3	304.5	384.3
Latin America	Peru	Finished stee	Consumption	.5	2730.5	2989.7	3178.3	2536.1	3730.7	2119.3	2459.3
Latin America	Trinidad	Finished stee	Consumption	.1	212.2	207.6	183.4	146.6	174.4	225.0	243.2
Latin America	Uruguay	Finished stee	Consumption	.1	202.8	211.3	191.6	234.2	279.2	292.9	252.4
Latin America	Venezuela	Finished stee	Consumption	.0	555.2	153.3	153.5	120.6	170.9	256.8	243.8
Latin America	Other Latin A	Finished stee	Consumption	.3	393.9	414.5	473.0	497.5	481.2	486.1	455.7
Europe	Albania	Finished stee	Consumption	.6	311.9	356.5	373.4	178.4	257.0	190.6	219.6
Europe	Austria	Finished stee	Consumption	.9	4455.8	4510.6	4313.9	3957.3	4871.0	4544.6	3463.4



Project Timeline

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation																								
Model development																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

Expected output

- Research Reports
- Models：H2-DRI-EAF production scheduling model with flexibility, etc.
- Research papers…

Thank You!